University of Pretoria

SEMESTER TEST: 15 MAY 2018

COURSE: Computer Science
PAPER: COS332

TIME: 1 hour
MARKS: 50

EXAMINER: Prof MS Olivier

THIS PAPER CONSISTS OF 10 PAGES.

ANSWER ALL QUESTIONS.
NO CALCULATORS PERMITTED

Question 1

AB STN BP

In each case select the alternative that fits the question best and write only the corresponding letter on your answer sheet.

1) The ISO OSI session layer is positioned (exactly) between the … and … layers.

   A: Network, Application

   B: Transport, Application

   C: Network, Presentation

   D: Transport, Presentation

   E: Presentation, Application

2) Some of the core Internet protocols were, amongst others, designed by … around 1980.

   A: Bill Gates

   B: Steve Wozniak

   C: Steve Jobs

   D: Clive Sinclair

   E: Vinton Cerf

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hour
: 50

3) A protocol is

A: A set of agreed-upon rules for message interchange.

B: Always described in an RFC.

C: Always open for interpretation.

D: Something that precedes a metacol.

E: A set of diagrams that indicate the sequence of messages that are interchanged.

4) A multicast packet 1110

A: Should be delivered to all subscribed parties.

B: In TCP/IP has an IP address with a first byte between 224 and 239, both included.

C: Is routed like a class C packet.

D: More than one of the above

E: All of the above

g

5) ARP translates a(n) IP ... address to a(n) ... Mac address.

A: IP, MAC

B: MAC, IP

C: TCP, IP

D: IPv4, IPv6

E: MAC, CAM

6) IP address 172.16.5.79 is a ... address.

A: Class A

B: Class B

C: Class C

D: Class D

E: Class E

7) How many class B IP network addresses can (potentially) exist? 11

A: ± 16 000

B: ± 64 000

C: ± 1 000 000

D: ± 4 000

E: ± 8 000

2^{14}

8) The TTL field in an IP datagram 1

2

A: Is used to encode a PING message.

B: Determines the destination port of the datagram.

C: Indicates whether the datagram has been fragmented.

D: Determines the maximum number of hops over which the packet can be forwarded.

E: None of the above

9) IP datagrams support source routing. Source routing is

A: Where the packet is routed back to the source.

B: Where all the packets from the same source are all sent together.

C: Where the source determines the route of the packet.

D: Frequently used on the Internet.

E: More than one of the above

10) IP fragmentation occurs because of

A: Different MTUs on the Internet.

B: Broken links.

C: A shortage of IP addresses.

D: Lost packets.

8, 8 = 4

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16 ×
8
128
h

11) An IPv6 address consists of

A: 128 bits. ✓

B: 16 octets. ✓

C: 8 hexadecimal numbers, separated by colons. ✓

D: More than one of the above

E: All of the above

12) ICMP is used to

A: Report errors that occur on the IP layer. ✓

B: Test network functions. ×

C: Modify routes. ×

d. D: More than one of the above ×

E: All of the above ×

13) Where are fragmented IP datagrams reassembled?

A: At the first router that notices that it is fragmented.

B: At the first router that can handle the total IP datagram.

C: At the final destination.

D: Whenever the TTL becomes 0.

E: Fragmented IP datagrams are discarded, rather than reassembled.

14) `ping` works by sending the following ICMP packet:

A: *Ping request*

B: *Echo request*

C: *Information request*

D: *Source quench*

E: *All of the above*

Recall Protocol

15) An ARP cache is maintained on

A: Each host connected to an Ethernet bus.

B: A special ARP cache server.

C: Layer 4 of the protocol stack.

D: The RARP server.

E: The hub.

16) What is the purpose of the netmask?

A: Allows a host to send frames to the router ✓

B: Used by routers to identify the network portion from the host portion of an IP address to enable proper routing of packets ✓

C: Used by administrators to join IP address blocks ✗

D: Used exclusively by routers to determine whether a frame must be forwarded or blocked ✓

17) A typical ARP message looks as follows:

A: Who has 10.1.1.1? Tell 10.1.1.2. ✓

B: Who has 00-04-38-68-da-00? Tell 00-11-43-e1-ba-4d.

C: `GET / HTTP/1.1`

D: What is the IP of xx.co.za?

E: What is the name of 196.7.147.37?

18) Suppose an entry in node A's routing table indicates that node B is the next node on the path to Z. Suppose an entry in node B's routing table indicates that the next hop on the path to Z is A. The resulting problem is addressed by using the following IP header field:

A: TTL

B: Source address

C: Destination address

D: Network unreachable

E: Been-there flag

Sliding window

19) Flow control in TCP uses

+ Selective

A: Stop-and-wait

B: The unrestricted approach

C: Window advertisements

D: RTS and CTS

20) If a TCP segment is lost, TCP reacts by

A: Retransmitting the segment.

B: Decreasing the rate at which segments are sent.

C: Performing another three-way handshake.

D: More than one of the above

E: All of the above

21) Which field does *not* occur in a UDP header?

A: Source port

B: Length

C: Sequence number

D: Checksum

22) A half-open attack occurs when

A: Only the first step of a three-way handshake is performed.

B: Only the first two steps of a three-way handshake are performed.

C: All three steps of a three-way handshake are performed.

D: A four-way handshake is performed.

E: No handshake is performed.

23) Which transport layer protocol(s) is/are used by DNS?

A: TCP

B: UDP

C: TCP and UDP

D: TCP/IP

E: IP

24) TCP packets are known as

A: Cells

B: Frames

C: Segments

D: Assemblies

E: Pipelines

25) During a slow start the amount of data sent will initially typically

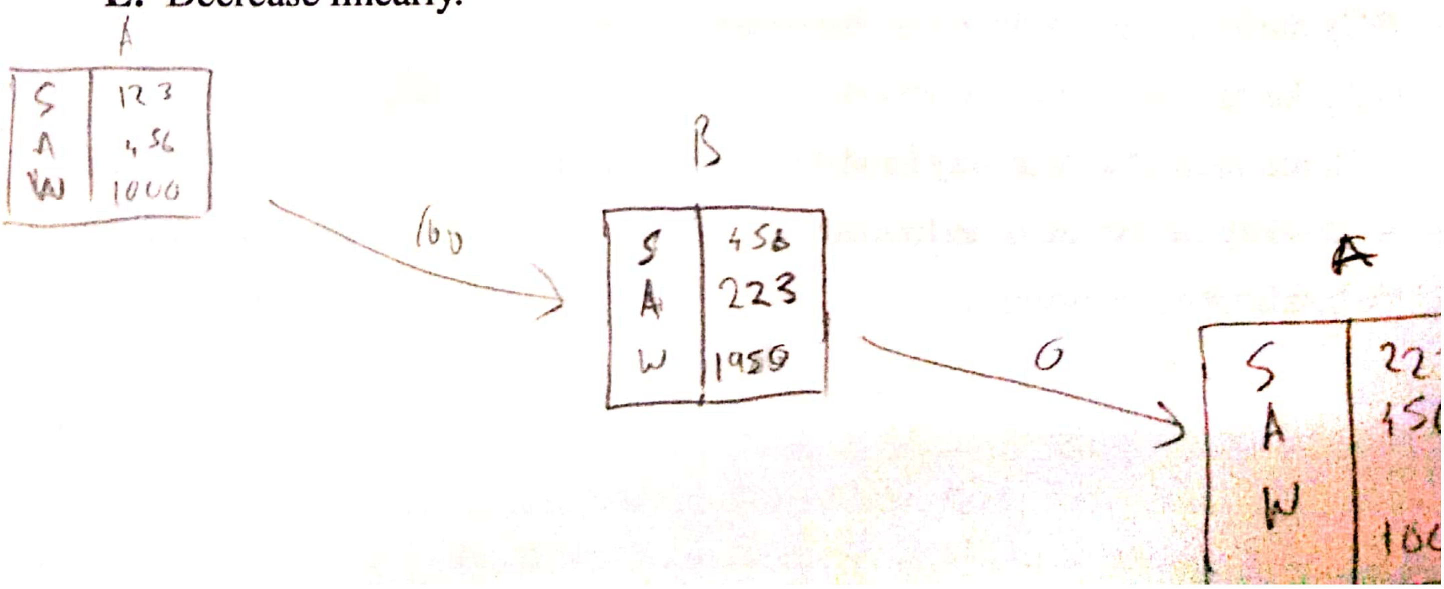
A: Remain constant.

B: Grow logarithmically.

C: Grow linearly.

D: Grow exponentially.

E: Decrease linearly.



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Question 2

a) Assume processes A and B are connected via a TCP connection. Assume that all data sent had been acknowledged. Assume that the variables at A are as follows (and that B knows this):

SEQ=123

ACK=456

WIN=1000

The window size at B is 2000

Now assume that the following events happen consecutively and slow enough that any message sent is received before any subsequent event happens:

a. A sends 100 bytes to B

b. B consumes 50 bytes

c. B sends 0 bytes to A

d. A sends 100 bytes to B

What are the values below, as included in the messages above?

(i) SEQ of message (c)

(ii) ACK of message (c)

(iii) Window size advertised in message (c)

(iv) SEQ of message (d)

(v) Window size advertised in message (d)

B

(5)

00 →

S	456
A	323
W	1856

1 2 4 8 16 32 64 128 256
9 1 2 3 4 5 6 7 8

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10

Question 2 (Continued)

00010000.000000
4+8+8 = 20

b) Consider the TCP header below.

```
 0                   1                   2                   3
 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|               ?               |               ?               |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|                        Sequence Number                        |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|                    Acknowledgement Number                     |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|  Data |           |U| |P|R| | |                               |
| Offset| Reserved  |R|?|S|S|?|?|            Window             |
|       |           |G| |H|T| | |                               |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|           Checksum            |         Urgent Pointer        |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|                    Options                    |    Padding    |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|                             data                              |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
```

512
64
576
16
592
8
600

Five of the fields have been masked with question marks. Write the names of those fields on your answer sheet. The order of the names is not important. (5)

Question 3 [10]

Assume an IP datagram I with destination address d arrives at a node N with IP address n. N has a routing table R. The netmask in use at N is m. Provide the five (5) steps of the routing algorithm that will be executed at N to route I. Use these identifiers as far as possible in your answer. For example, the test $r \neq n$ (which may or may not be part of the routing algorithm) is clear without any further explanation. However, to return the packet to the sender (if such an action forms part of the routing algorithm) cannot be expressed using this notation. Hence a typical step in the algorithm may be succinctly written as

$$\text{if } n \neq d \text{ then return } I \text{ to its sender}$$

Write each of the steps on your answer sheet next to the relevant step number (1 to 5) provided on your answer sheet. [5]

16+8+4+2+1

172.0001 1111.1111.11111 100.00000

172 . . 4 . 0

255

96 +
8
104

172
255 -
221
034

Question 4

As you probably know, RFC1918 reserves the IPv4 address blocks 10.0.0.0/8, 172.16.0.0/12 and 192.168.0.0/16 (in the traditional A, B and C class address spaces, respectively) for private use. Assume ICANN decides that the private addresses in the class B network space are hardly ever used for their intended purpose, and decides to make these addresses available as public addresses. For technical reason they decide to split the block into at least 1000 blocks (or networks) such that each network can address at least 1000 hosts.

Assume that they subnet the current block into smaller blocks (or networks) to achieve these goals. Let A be the new network with the smallest network address and Z be the new network with the largest network address. (Remember that they want *at least* 1000 networks, but the scheme they use may result in more than 1000 networks.)

a) What is the network address of network A? Use CIDR notation. (2)

b) What is the network address of network Z? Use CIDR notation. (2)

c) What is the address range network Z? In other words, what are the

  (i) Smallest; and

  (ii) Largest

  IPv4 addresses that can be assigned to hosts in network Z? (2)

d) What is the broadcast address for network A? (1)

e) What is the broadcast address for network Z? (1)

f) Consider a computer with the address 172.25.109.221.

  (i) In which subnetwork will this computer be located? (Your answer will be a number from 0 to 999 — or more.)

  (ii) What will host number will this computer have in this subnetwork? (Tour answer will be a number from 1 to 1000 — or more.) (2)

.00011001.01101101.11011101

[10]

TOTAL

1001011011 0111011101

512
64
256 +

[50]

END OF PAPER

Question 1

1	D ✓	14	B ✓
2	E ✓	15	A ✓
3	A ✓	16	B ✓
4	D ✓	17	A ✓
5	A ✓	18	A ✓
6	B ✓	19	D C
7	B A	20	D ✓
8	D ✓	21	C ✓
9	C ✓	22	A B
10	D A	23	B C
11	E ✓	24	C ✓
12	D A	25	D ✓
13	C ✓	—	—

19/25

Question 2

a)(i)	456 ✓
(ii)	223 ✓
(iii)	1950 ✓
(iv)	223 ✓
(v)	1000 ✓
b)•	Source Port ✓
•	Destination Port ✓
•	ACK ✓
•	SYN ✓
•	FIN ✓

10/10

Question 3

1)	netAddress = d && m OK ...
2)	if (netAddress == netAdress(N))
	send I to host (same network)
3)	else if (netAddress ∈ R)
	send I to router at netAddress ?
4)	else discard I
5)	

3 /5

Question 4

a)	172.16.4.0 / 22
b)	172.31.252.0 / 22
c)(i)	172.31.252.1
c)(ii)	172.31.255.254
d)	172.16.7.255
e)	172.31.255.255
f)(i)	603
f)(ii)	477

8 /10

University of Pretoria

8 JUNE 2016

COURSE: Computer Science
PAPER: COS332

TIME: 3 hours
MARKS: 100

EXAMINER: 1. Prof MS Olivier
2. Dr DT van der Haar (UJ)

THIS PAPER CONSISTS OF 17 PAGES.

ANSWER ALL QUESTIONS.
NO CALCULATORS PERMITTED.
AN ANSWER SHEET AND ADDENDUM ARE PROVIDED SEPARATELY.
WHEN A QUESTION CAN BE ANSWERED BY MEANS OF AN ACRONYM IT IS NOT NECESSARY TO PROVIDE THE PHRASE THAT THE ACRONYM REPRESENTS.
STUDENTS ARE ALLOWED TO WRITE ON THE EXAMINATION PAPER.
THE TOTAL NUMBER OF MARKS AVAILABLE IN THE PAPER IS 101; YOU MAY ANSWER ANY QUESTIONS THAT ADD UP TO 100 MARKS (OR LESS).

Question 1

In each case select the alternative that fits the question best and write only the corresponding letter on your answer sheet.

1) Manchester encoding is better than NRZ because

   A: Manchester is self-synchronising.

   B: The baud rate of Manchester is higher than that of NRZ.

   C: The baud rate of Manchester is lower than that of NRZ.

   D: The wires may be switched in the case of Manchester.

   E: None of the above

Question 1 (Continued)

2) A switch (used in the place of a hub)

   A: Connects different networks and can convert protocols.

   B: Makes a connection only between computers that are busy communicating.

   C: Routes packets between nodes.

   D: Operates on ISO OSI layer 3.

   E: More than one of the above

3) An E1 line communicates at

   A: the same speed as a T1 line

   B: 1.5 Mbps

   C: 2.048 Mbps

   D: 10.000 Mbps

   E: All of the above

4) Consider ADSL. The 'A' signifies that it is asymmetric; in other words

   A: The upload and download speeds are different

   B: The electrical potential of one line is varied while the other is grounded.

   C: It establishes a connection between a user and an ISP.

   D: It differs from one country to another.

   E: It is not fully digital.

5) Which of the following countries is *not* directly linked to SEACOM?

   A: South Africa

   B: Mozambique

   C: DRC

   D: Egypt

   E: India

Question 1 (Continued)

6) Which of the following is (are) *not* a function of ISO/OSI's layer 2?

A: Media access control
B: Data delineation
C: Dialogue control
D: Error control
E: More than one of the above

7) A header added to a message at ISO OSI layer n will be removed at the destination at layer . . .

A: $n - 1$
B: n
C: $n + 1$
D: 1
E: 7

8) Layers 1 to 4 of the ISO OSI model are known as the

A: Network-oriented layers
B: Application-oriented layers
C: TCP/IP-oriented layers
D: More than one of the above
E: None of the above

9) If parity is used for error control, only errors that affect . . . will be detected.

A: A single bit
B: An odd number of bits
C: An even number of bits
D: All of the above
E: None of the above

Question 1 (Continued)

10) A protocol is

A: A set of agreed-upon rules for message interchange.
B: Always described in an RFC.
C: Always open for interpretation.
D: Something that precedes a metacol.
E: A set of diagrams that indicate the sequence of messages that are interchanged.

11) The `za` ccTLD is controlled by

A: AfriNIC
B: ZADNA
C: `.za`
D: Uniforum
E: `.co.za`

12) `bind` is a program that provides . . . services.

A: directory
B: address
C: routing
D: name
E: None of the above

13) The `whois` command should (eventually) query the . . . records.

A: ICANN
B: IANA
C: Registrar's
D: TLD registry
E: `who.is`

Question 1 (Continued)

14) Which IEEE standard specifies the operation of wireless local area networks?

A: IEEE 802.3

B: IEEE 802.5

C: IEEE 802.10

D: IEEE 802.11

15) The Ethernet address `FF:FF:FF:FF:FF:FF` is

A: A MAC address

B: A Broadcast address

C: A Group address

D: Invalid

E: None of the above

16) An Ethernet frame may contain ... octets of data.

A: 0 – 1500

B: $46 - \infty$

C: 46 – 1500

D: 2 – 1500

E: 150 – 4800

17) When no monitor's presence is detected on an 802.5 network

A: The hub appoints a new monitor.

B: The previous monitor resumes its duties.

C: The station with the highest priority starts acting as monitor.

D: A new monitor is elected by the remaining nodes.

E: The network simply continues operating because it does not really need a monitor.

Question 1 (Continued)

18) Suppose a node using SDLC has to transmit the data `01111110`. The data will be observed on the line as

A: `011111010`

B: `01111110`

C: `011111100`

D: `011111110`

E: `011111X10`

19) ARP translates a(n) ... address to a(n) ... address.

A: IP, MAC

B: MAC, IP

C: TCP, IP

D: IPv4, IPv6

E: MAC, CAM

20) Suppose the first octet of a UTF-8 character is 193. How long (in octets) will the entire character be?

A: 1

B: 2

C: 3

D: 4

E: 5

F: 6

Question 1 (Continued)

21) On which layer is SSL/TLS usually positioned?

A: 6

B: 5

C: 'Just above 5'

D: 'Just above 4'

E: 4

22) Huffman code is used to

A: Compress data

B: Correct errors in data

C: Encrypt data

D: More than one of the above

E: All of the above

23) To use POP3 anonymously, one uses the command

A: USER ANON

B: USER ANONYMOUS

C: NOUSER

D: USER *

E: None of the above

24) IP address 172.16.5.79 is a … address.

A: Class A

B: Class B

C: Class C

D: Class D

E: Class E

Question 1 (Continued)

25) IP fragmentation occurs because of

A: Different MTUs on the Internet.

B: Broken links.

C: A shortage of IP addresses.

D: Lost packets.

26) ICMP is used to

A: Report errors that occur on the IP layer.

B: Test network functions.

C: Modify routes.

D: More than one of the above

E: All of the above

27) `ping` works by sending the following ICMP packet:

Logic

A: *Ping request*

B: *Echo request*

C: *Information request*

D: *Source quench*

E: All of the above

28) Which of the following IP addresses does *not* belong to a private IP address block?

A: 10.11.12.13

B: 172.16.17.18

C: 192.168.169.170

D: 238.239.240.241

Question 1 (Continued)

29) What is the purpose of the netmask?

A: Allows a host to send frames to the router

B: Used by routers to identify the network portion from the host portion of an IP address to enable proper routing of packets

C: Used by administrators to join IP address blocks

D: Used exclusively by routers to determine whether a frame must be forwarded or blocked

30) A typical ARP message looks as follows:

A: Who has 10.1.1.1? Tell 10.1.1.2.

B: Who has 00-04-38-68-da-00? Tell 00-11-43-e1-ba-4d.

C: `GET / HTTP/1.1`

D: What is the IP of xx.co.za?

E: What is the name of 196.7.147.37?

31) An AS (Autonomous System) is usually a

A: Personal computer.

B: Network controlled by a single authority.

C: Mainframe computer.

D: ccTLD.

E: DNS registrar.

32) Which protocol has reliability as one of its main goals?

A: IP

B: TCP

C: UDP

D: More than one of the above

E: All of the above

Question 1 (Continued)

33) During a slow start the amount of data sent will initially typically

A: Remain constant.

B: Grow logarithmically.

C: Grow linearly.

D: Grow exponentially.

E: Decrease linearly.

34) Which of the following considerations is the most important for BGP *(Border Gateway Protocol)*?

A: Number of hops

B: Nature of routes inside organisations

C: Routing policies

D: Dijkstra's algorithm

E: ICMP information about network conditions

35) Consider the problem of *transparency* in SMTP. Which of the following is true?

A: Transparency is not an issue — any type of data can be e-mailed.

B: In most e-mail programs users should avoid typing a full stop on a line of its own, because that will prematurely terminate the body of the message and cause only the part prior to the full stop to be sent.

C: POP3 solves the SMTP transparency problem.

D: IMAP solves the SMTP transparency problem.

E: None of the above

Question 1 (Continued)

36) Which of the following is a valid DNS root server?

A: f.root-servers.net

B: .com

C: p.root-servers.net

D: f.root-server.net

E: More than one of the above

37) If SMTP cannot deliver a message, it may

A: Wait and try again later.

B: Send the message to a secondary server.

C: Return an error message to the sender.

D: More than one of the above

E: All of the above

38) An application protocol will typically respond with a message in the ...-range when security or configuration errors prevented it from completing a request.

A: 100s

B: 200s

C: 300s

D: 400s

E: 500s

39) The most important routing protocol at JINX is

A: RIP

B: BGP

C: OSPF

D: Dijkstra

E: Bellman-Ford

Question 1 (Continued)

40) A multicast packet

A: Should be delivered to all subscribed parties.

B: In TCP/IP has an IP address with a first byte between 224 and 239, both included.

C: Is routed like a class C packet.

D: More than one of the above

E: All of the above

[40]

Question 2

a) With which well-known ports are the following services usually associated?

(i) HTTP 80
(ii) DNS 53
(iii) Telnet 23
(iv) SSH 21
(v) SMTP 25 (5)

b) What are the following ports typically used for? (It is not necessary to give the corresponding port numbers.)

(i) Well-known ports Applications
(ii) Registered ports PVT Applications
(iii) Dynamic and/or Private Ports Testing (3)

c) Why are ports useful in firewalls? for routing data (1)

d) How many bits are used to identify a port in TCP/IP? 16 (1)

[10]

Question 3

Assume some computer X has one network interface with MAC address `10-20-30-40-50-60` and IPv4 address 10.20.30.40. Assume, for the purposes of this question that 10.0.0.0/8 is a routable, public network address. Whenever a question below refers to an IP address (or even just an address) IPv6 is implied.

a) What is the loopback address of X? fe::/80 (1)

b) What unspecified address will X use whenever necessary? 1020:30FF:F… (1)

c) What is the link-local address of X? No privacy-protection mechanisms are to be used. Fe::/8 (2)

d) What is the IPv4 mapped address of X? 10.20... (2)

e) What would the IPv4 compatible address of X have been? (2)

f) If X wants to use 6to4 tunnelling, what would its network address be? (Use CIDR-notation.) B (3)

g) If X were to use Teredo tunnelling, what prefix would indicate that is was using such tunnelling? (2)

h) What (destination) prefix would X use when it transmits a multicast packet? 0001 (2)

[15]

Question 4

The addendum contains copies of 13 packets captured with Wireshark on a spanning port of a managed switch. These 13 packets are in the sequence in which they have been captured. The capture lasted for only a few seconds; these 13 packets are just a few of the thousands of packets that were caputured during this period. Packets that are not relevant to this question have been omiitted. Some details of these 13 packets that are not relevant to this question have been hidden or omitted. Line numbers have been added to the contents of each packet for ease of reference. Where line numbers are missing it indicates that one or more lines have been removed because they were not deemed relevant. Answer the questions below about regarding the packets contained in the addendum.

a) Consider packet 1.

(i) Which application layer protocol is used? IPV4 , UDP , E

(ii) Which transport layer protocol is used?

(iii) Which network layer protocol is used?

(iv) which data link layer protocol is used? (4)

b) Details about a number of network interfaces have been captured. What are the vendor IDs of interfaces in these packets of the interfaces supplied by each of the following manufacturers?

(i) ASRock

(ii) Raspberry Pi

(iii) TP-Link (3)

c) Consider packets 1 and 2.

(i) Explain the action of node 10.1.1.31. Why does it transmit packet 2?

(ii) The TTL fields of packets 1 and 2 are quite different (128 vs 64). What is the (most likely) explanation for this difference?

(iii) The destination ports of packets 1 and 2 are the same (53). Why?

(iv) The source ports of packets 1 and 2 are quite different (63933 and 8312, respectively). Briefly explain this phenomenon. (1)

(4)

Question 4 (Continued)

d) Consider packet 3. Assume that it is a response to packet 2.

(i) What information on layer 4 supports that packet 3 is a response to packet 2?

(ii) What information on layer 3 may support that packet 3 is a response to packet 2?

(iii) What information on layer 2 may support that packet 3 is a response to packet 2?

(iv) Is the application layer response in packet 3 iterative or recursive? Why? (4)

e) Consider packet 4.

(i) All the addresses, as well as the destination ports in packets 2 and 4 are identical. Briefly explain why the source ports are not.

(ii) The TTL fields of packets 2 and 4 are also identical (64). Would one expect this, or is it coincidental? Explain briefly.

(iii) It seems that it was not necessary for 10.1.1.31 to send this packet. Why?

(iv) Why do you think 10.1.1.31 sent this packet even though it was not really necessary? (4)

f) Consider packet 6. To which of the packets provided in the addendum is it (most likely) a response? (1)

g) Consider packet 7. In what way does it differ from packet 1 on the application layer? (1)

h) Consider packet 9.

(i) The trace contained no response to the query issued in packet 8. Given this fact, what feature of the DNS does packet 9 exploit? Briefly explain.

(ii) While packets 8 and 9 are very similar, they are sent to different destinations (8.8.8.8 and 192.5.4.1, respectively). Explain why both packets use the same destination address on layer 2. (2)

Question 4 (Continued)

i) Assume packet 10 is a response to packet 9.

(i) Packet 10 does not contain the answer to the query contained in packet 9. Justify this claim.

(ii) What type of information is provided as 'answer' in packet 10?

(iii) What should one conclude from the fact that a proper answr is not provided?

(iv) Is the IPv6 address provided in line 53 of packet 10 somehow derived from an IPv4 address? Why (not)? (4)

j) Consider the merit of sending packet 11 at this time.

(i) Why may packet 11 provide useful information?

(ii) Why may packet 11 be problematic? (2)

[26]

Question 5

Consider the packets captured on a link between two hosts that are provided in the addendum and marked packet A through packet J. (See the previous question about more details about how they were captured.) Answer the questions below based on these traces. For ease of reference you may assume that 10.1.1.15 is computer X and 54.164.213.10 is computer Y.

a) What is the purpose of packets A, B and C? Justify your answer. (1)

b) Consider packet D.

(i) What is its application layer purpose?

(ii) What is `squid` (line 41)?

(iii) The TCP ACK flag is set. What is acknowledged? (3)

Question 5 (Continued)

c) Packet E does not carry any payload — its length is 0.

(i) For what purpose was E sent?

(ii) The TCP `Seq` field has the value 1. Is this correct? Why (not)?

(iii) The TCP `Ack` field has the value 618. Is this correct? Why (not)?

(iv) The calculated window size advertisement is 15872. Is this (approximately) what you expected? Why (not)? (4)

d) What is the MIME type of the payload in packet F? (1)

e) Which node (X or Y) initiates termination of the connection? Justify your answer. (1)

[10]

TOTAL **[100]**

END OF PAPER

Packet 3

```
Frame 299052: 124 bytes on wire (992 bits), 124 bytes captured (992 bits) on ir
Ethernet II, Src: Tp-LinkT_e7:76:46 (30:b5:c2:e7:76:46), Dst: Raspberr_ca:1f:1e
    Destination: Raspberr_ca:1f:1e (b8:27:eb:ca:1f:1e)
    Source: Tp-LinkT_e7:76:46 (30:b5:c2:e7:76:46)
    Type: IPv4 (0x0800)
Internet Protocol Version 4, Src: 8.8.8.8, Dst: 10.1.1.31
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes
    Differentiated Services Field: 0x0c (DSCP: Unknown, ECN: Not-ECT)
    Total Length: 110
    Identification: 0x15ef (5615)
    Flags: 0x00
    Fragment offset: 0
    Time to live: 55
    Protocol: UDP (17)
    Header checksum: 0x5255 [validation disabled]
    Source: 8.8.8.8
    Destination: 10.1.1.31
    [Source GeoIP: Unknown]
    [Destination GeoIP: Unknown]
User Datagram Protocol, Src Port: 53 (53), Dst Port: 8312 (8312)
    Source Port: 53
    Destination Port: 8312
    Length: 90
    Checksum: 0xaaae [validation disabled]
    [Stream index: 338]
Domain Name System (response)
    [Request In: 297958]
    [Time: 0.276987000 seconds]
    Transaction ID: 0x80b0
    Flags: 0x8190 Standard query response, No error
    Questions: 1
    Answer RRs: 2
    Authority RRs: 0
    Additional RRs: 1
    Queries
        mail.cs.up.ac.za: type A, class IN
            Name: mail.cs.up.ac.za
            [Name Length: 16]
            [Label Count: 5]
            Type: A (Host Address) (1)
            Class: IN (0x0001)
    Answers
        mail.cs.up.ac.za: type CNAME, class IN, cname hermes.cs.up.ac.za
        hermes.cs.up.ac.za: type A, class IN, addr 137.215.40.239
```

Packet 4

```
Frame 299074: 89 bytes on wire (712 bits), 89 bytes captured (712 bits) on inte
Ethernet II, Src: Raspberr_ca:1f:1e (b8:27:eb:ca:1f:1e), Dst: Tp-LinkT_e7:76:4
    Destination: Tp-LinkT_e7:76:46 (30:b5:c2:e7:76:46)
    Source: Raspberr_ca:1f:1e (b8:27:eb:ca:1f:1e)
    Type: IPv4 (0x0800)
Internet Protocol Version 4, Src: 10.1.1.31, Dst: 8.8.8.8
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes
    Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 75
    Identification: 0xbee7 (48871)
    Flags: 0x00
    Fragment offset: 0
    Time to live: 64
    Protocol: UDP (17)
    Header checksum: 0xa08b [validation disabled]
    Source: 10.1.1.31
    Destination: 8.8.8.8
    [Source GeoIP: Unknown]
    [Destination GeoIP: Unknown]
User Datagram Protocol, Src Port: 8535 (8535), Dst Port: 53 (53)
    Source Port: 8535
    Destination Port: 53
    Length: 55
    Checksum: 0x902e [validation disabled]
    [Stream index: 339]
Domain Name System (query)
    Transaction ID: 0x20e7
    Flags: 0x0110 Standard query
    Questions: 1
    Answer RRs: 0
    Authority RRs: 0
    Additional RRs: 1
    Queries
        hermes.cs.up.ac.za: type A, class IN
            Name: hermes.cs.up.ac.za
            [Name Length: 18]
            [Label Count: 5]
            Type: A (Host Address) (1)
            Class: IN (0x0001)
```

Packet 5

```
Frame 300334: 105 bytes on wire (840 bits), 105 bytes captured (840 bits) on ir
Ethernet II, Src: Tp-LinkT_e7:76:46 (30:b5:c2:e7:76:46), Dst: Raspberr_ca:1f:1e
    Destination: Raspberr_ca:1f:1e (b8:27:eb:ca:1f:1e)
    Source: Tp-LinkT_e7:76:46 (30:b5:c2:e7:76:46)
    Type: IPv4 (0x0800)
Internet Protocol Version 4, Src: 8.8.8.8, Dst: 10.1.1.31
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes
    Differentiated Services Field: 0x0c (DSCP: Unknown, ECN: Not-ECT)
    Total Length: 91
    Identification: 0xcb07 (51975)
    Flags: 0x00
    Fragment offset: 0
    Time to live: 55
    Protocol: UDP (17)
    Header checksum: 0x9d4f [validation disabled]
    Source: 8.8.8.8
    Destination: 10.1.1.31
    [Source GeoIP: Unknown]
    [Destination GeoIP: Unknown]
User Datagram Protocol, Src Port: 53 (53), Dst Port: 8535 (8535)
    Source Port: 53
    Destination Port: 8535
    Length: 71
    Checksum: 0x9812 [validation disabled]
    [Stream index: 339]
Domain Name System (response)
    [Request In: 299075]
    [Time: 0.487202000 seconds]
    Transaction ID: 0x20e7
    Flags: 0x8190 Standard query response, No error
    Questions: 1
    Answer RRs: 1
    Authority RRs: 0
    Additional RRs: 1
    Queries
        hermes.cs.up.ac.za: type A, class IN
            Name: hermes.cs.up.ac.za
            [Name Length: 18]
            [Label Count: 5]
            Type: A (Host Address) (1)
            Class: IN (0x0001)
    Answers
        hermes.cs.up.ac.za: type A, class IN, addr 137.215.40.239
```

Packet 6

```
Frame 300334: 105 bytes on wire (840 bits), 105 bytes captured (840 bits) on ir
Ethernet II, Src: Tp-LinkT_e7:76:46 (30:b5:c2:e7:76:46), Dst: Raspberr_ca:1f:1e
    Destination: Raspberr_ca:1f:1e (b8:27:eb:ca:1f:1e)
    Source: Tp-LinkT_e7:76:46 (30:b5:c2:e7:76:46)
    Type: IPv4 (0x0800)
Internet Protocol Version 4, Src: 8.8.8.8, Dst: 10.1.1.31
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes
    Differentiated Services Field: 0x0c (DSCP: Unknown, ECN: Not-ECT)
    Total Length: 91
    Identification: 0xcb07 (51975)
    Flags: 0x00
    Fragment offset: 0
    Time to live: 55
    Protocol: UDP (17)
    Header checksum: 0x9d4f [validation disabled]
    Source: 8.8.8.8
    Destination: 10.1.1.31
    [Source GeoIP: Unknown]
    [Destination GeoIP: Unknown]
User Datagram Protocol, Src Port: 53 (53), Dst Port: 8535 (8535)
    Source Port: 53
    Destination Port: 8535
    Length: 71
    Checksum: 0x9812 [validation disabled]
    [Stream index: 339]
Domain Name System (response)
    [Request In: 299075]
    [Time: 0.487202000 seconds]
    Transaction ID: 0x20e7
    Flags: 0x8190 Standard query response, No error
    Questions: 1
    Answer RRs: 1
    Authority RRs: 0
    Additional RRs: 1
    Queries
        hermes.cs.up.ac.za: type A, class IN
            Name: hermes.cs.up.ac.za
            [Name Length: 18]
            [Label Count: 5]
            Type: A (Host Address) (1)
            Class: IN (0x0001)
    Answers
        hermes.cs.up.ac.za: type A, class IN, addr 137.215.40.239
```

Packet 7

```
Frame 300357: 76 bytes on wire (608 bits), 76 bytes captured (608 bits) on inte
Ethernet II, Src: AsrockIn_29:d3:3e (d0:50:99:29:d3:3e), Dst: Raspberr_ca:1f:1e
    Destination: Raspberr_ca:1f:1e (b8:27:eb:ca:1f:1e)
    Source: AsrockIn_29:d3:3e (d0:50:99:29:d3:3e)
    Type: IPv4 (0x0800)
Internet Protocol Version 4, Src: 10.1.1.2, Dst: 10.1.1.31
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes
    Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 62
    Identification: 0x349e (13470)
    Flags: 0x00
    Fragment offset: 0
    Time to live: 128
    Protocol: UDP (17)
    Header checksum: 0xefee [validation disabled]
    Source: 10.1.1.2
    Destination: 10.1.1.31
    [Source GeoIP: Unknown]
    [Destination GeoIP: Unknown]
User Datagram Protocol, Src Port: 63934 (63934), Dst Port: 53 (53)
    Source Port: 63934
    Destination Port: 53
    Length: 42
    Checksum: 0x81b1 [validation disabled]
    [Stream index: 340]
Domain Name System (query)
    Transaction ID: 0x0013
    Flags: 0x0100 Standard query
    Questions: 1
    Answer RRs: 0
    Authority RRs: 0
    Additional RRs: 0
    Queries
        mail.cs.up.ac.za: type AAAA, class IN
            Name: mail.cs.up.ac.za
            [Name Length: 16]
            [Label Count: 5]
            Type: AAAA (IPv6 Address) (28)
            Class: IN (0x0001)
```

Packet 8

```
Frame 300371: 89 bytes on wire (712 bits), 89 bytes captured (712 bits) on inte
Ethernet II, Src: Raspberr_ca:1f:1e (b8:27:eb:ca:1f:1e), Dst: Tp-LinkT_e7:76:4
    Destination: Tp-LinkT_e7:76:46 (30:b5:c2:e7:76:46)
    Source: Raspberr_ca:1f:1e (b8:27:eb:ca:1f:1e)
    Type: IPv4 (0x0800)
Internet Protocol Version 4, Src: 10.1.1.31, Dst: 8.8.8.8
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes
    Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 75
    Identification: 0xbf02 (48898)
    Flags: 0x00
    Fragment offset: 0
    Time to live: 64
    Protocol: UDP (17)
    Header checksum: 0xa070 [validation disabled]
    Source: 10.1.1.31
    Destination: 8.8.8.8
    [Source GeoIP: Unknown]
    [Destination GeoIP: Unknown]
User Datagram Protocol, Src Port: 9533 (9533), Dst Port: 53 (53)
    Source Port: 9533
    Destination Port: 53
    Length: 55
    Checksum: 0x784f [validation disabled]
    [Stream index: 341]
Domain Name System (query)
    Transaction ID: 0x34c5
    Flags: 0x0110 Standard query
    Questions: 1
    Answer RRs: 0
    Authority RRs: 0
    Additional RRs: 1
    Queries
        hermes.cs.up.ac.za: type AAAA, class IN
            Name: hermes.cs.up.ac.za
            [Name Length: 18]
            [Label Count: 5]
            Type: AAAA (IPv6 Address) (28)
            Class: IN (0x0001)
```

Packet 9

```
Frame 303165: 89 bytes on wire (712 bits), 89 bytes captured (712 bits) on inte
Ethernet II, Src: Raspberr_ca:1f:1e (b8:27:eb:ca:1f:1e), Dst: Tp-LinkT_e7:76:4(
    Destination: Tp-LinkT_e7:76:46 (30:b5:c2:e7:76:46)
    Source: Raspberr_ca:1f:1e (b8:27:eb:ca:1f:1e)
    Type: IPv4 (0x0800)
Internet Protocol Version 4, Src: 10.1.1.31, Dst: 192.5.4.1
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes
    Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 75
    Identification: 0x449d (17565)
    Flags: 0x00
    Fragment offset: 0
    Time to live: 64
    Protocol: UDP (17)
    Header checksum: 0x66df [validation disabled]
    Source: 10.1.1.31
    Destination: 192.5.4.1
    [Source GeoIP: Unknown]
    [Destination GeoIP: Unknown]
User Datagram Protocol, Src Port: 54556 (54556), Dst Port: 53 (53)
    Source Port: 54556
    Destination Port: 53
    Length: 55
    Checksum: 0x1cd3 [validation disabled]
    [Stream index: 344]
Domain Name System (query)
    [Response In: 303964]
    Transaction ID: 0x2d6b
    Flags: 0x0010 Standard query
    Questions: 1
    Answer RRs: 0
    Authority RRs: 0
    Additional RRs: 1
    Queries
        hermes.cs.up.ac.za: type AAAA, class IN
            Name: hermes.cs.up.ac.za
            [Name Length: 18]
            [Label Count: 5]
            Type: AAAA (IPv6 Address) (28)
            Class: IN (0x0001)
```

Packet 10

```
1 Frame 303963: 302 bytes on wire (2416 bits), 302 bytes captured (2416 bits) on
2 Ethernet II, Src: Tp-LinkT_e7:76:46 (30:b5:c2:e7:76:46), Dst: Raspberr_ca:1f:1e
3     Destination: Raspberr_ca:1f:1e (b8:27:eb:ca:1f:1e)
4     Source: Tp-LinkT_e7:76:46 (30:b5:c2:e7:76:46)
5     Type: IPv4 (0x0800)
6 Internet Protocol Version 4, Src: 192.5.4.1, Dst: 10.1.1.31
7     0100 .... = Version: 4
8     .... 0101 = Header Length: 20 bytes
9     Differentiated Services Field: 0x10 (DSCP: Unknown, ECN: Not-ECT)
10    Total Length: 288
11    Identification: 0x4375 (17269)
12    Flags: 0x00
13    Fragment offset: 0
14    Time to live: 55
15    Protocol: UDP (17)
16    Header checksum: 0x7022 [validation disabled]
17    Source: 192.5.4.1
18    Destination: 10.1.1.31
19    [Source GeoIP: Unknown]
20    [Destination GeoIP: Unknown]
21 User Datagram Protocol, Src Port: 53 (53), Dst Port: 54556 (54556)
22    Source Port: 53
23    Destination Port: 54556
24    Length: 268
27 Domain Name System (response)
28    [Request In: 303165]
29    [Time: 0.335626000 seconds]
30    Transaction ID: 0x2d6b
31    Flags: 0x8010 Standard query response, No error
32    Questions: 1
33    Answer RRs: 0
34    Authority RRs: 4
35    Additional RRs: 6
36    Queries
37        hermes.cs.up.ac.za: type AAAA, class IN
38            Name: hermes.cs.up.ac.za
39            [Name Length: 18]
40            [Label Count: 5]
41            Type: AAAA (IPv6 Address) (28)
42            Class: IN (0x0001)
43    Authoritative nameservers
44        ac.za: type NS, class IN, ns za-ns.anycast.pch.net
45        ac.za: type NS, class IN, ns ucthpx.uct.ac.za
46        ac.za: type NS, class IN, ns disa.tenet.ac.za
47        ac.za: type NS, class IN, ns ns0.is.co.za
48    Additional records
49        ns0.is.co.za: type A, class IN, addr 196.4.160.17
50        disa.tenet.ac.za: type A, class IN, addr 196.21.79.50
51        disa.tenet.ac.za: type AAAA, class IN, addr 2001:4200:ffff:a::1
52        ucthpx.uct.ac.za: type A, class IN, addr 137.158.128.1
53        ucthpx.uct.ac.za: type AAAA, class IN, addr 2001:43f8:75::3
```

Packet 11

```
Frame 303972: 89 bytes on wire (712 bits), 89 bytes captured (712 bits) on inte
Ethernet II, Src: Raspberr_ca:1f:1e (b8:27:eb:ca:1f:1e), Dst: Tp-LinkT_e7:76:46
    Destination: Tp-LinkT_e7:76:46 (30:b5:c2:e7:76:46)
    Source: Raspberr_ca:1f:1e (b8:27:eb:ca:1f:1e)
    Type: IPv4 (0x0800)
Internet Protocol Version 4, Src: 10.1.1.31, Dst: 8.8.8.8
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes
    Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 75
    Identification: 0xbf0b (48907)
    Flags: 0x00
    Fragment offset: 0
    Time to live: 64
    Protocol: UDP (17)
    Header checksum: 0xa067 [validation disabled]
    Source: 10.1.1.31
    Destination: 8.8.8.8
    [Source GeoIP: Unknown]
    [Destination GeoIP: Unknown]
User Datagram Protocol, Src Port: 37222 (37222), Dst Port: 53 (53)
    Source Port: 37222
    Destination Port: 53
    Length: 55
    Checksum: 0x0183 [validation disabled]
    [Stream index: 346]
Domain Name System (query)
    Transaction ID: 0x3f68
    Flags: 0x0110 Standard query
    Questions: 1
    Answer RRs: 0
    Authority RRs: 0
    Additional RRs: 1
    Queries
        hermes.cs.up.ac.za: type AAAA, class IN
            Name: hermes.cs.up.ac.za
            [Name Length: 18]
            [Label Count: 5]
            Type: AAAA (IPv6 Address) (28)
            Class: IN (0x0001)
```

Packet 12

```
Frame 304879: 138 bytes on wire (1104 bits), 138 bytes captured (1104 bits) on
Ethernet II, Src: Tp-LinkT_e7:76:46 (30:b5:c2:e7:76:46), Dst: Raspberr_ca:1f:1e
    Destination: Raspberr_ca:1f:1e (b8:27:eb:ca:1f:1e)
    Source: Tp-LinkT_e7:76:46 (30:b5:c2:e7:76:46)
    Type: IPv4 (0x0800)
Internet Protocol Version 4, Src: 8.8.8.8, Dst: 10.1.1.31
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes
    Differentiated Services Field: 0x0c (DSCP: Unknown, ECN: Not-ECT)
    Total Length: 124
    Identification: 0x2418 (9240)
    Flags: 0x00
    Fragment offset: 0
    Time to live: 55
    Protocol: UDP (17)
    Header checksum: 0x441e [validation disabled]
    Source: 8.8.8.8
    Destination: 10.1.1.31
    [Source GeoIP: Unknown]
    [Destination GeoIP: Unknown]
User Datagram Protocol, Src Port: 53 (53), Dst Port: 37222 (37222)
    Source Port: 53
    Destination Port: 37222
    Length: 104
    Checksum: 0x5def [validation disabled]
    [Stream index: 346]
Domain Name System (response)
    [Request In: 303973]
    [Time: 0.403078000 seconds]
    Transaction ID: 0x3f68
    Flags: 0x8190 Standard query response, No error
    Questions: 1
    Answer RRs: 0
    Authority RRs: 1
    Additional RRs: 1
    Queries
        hermes.cs.up.ac.za: type AAAA, class IN
            Name: hermes.cs.up.ac.za
            [Name Length: 18]
            [Label Count: 5]
            Type: AAAA (IPv6 Address) (28)
            Class: IN (0x0001)
    Authoritative nameservers
        cs.up.ac.za: type SOA, class IN, mname ns2.up.ac.za
```

Packet 13

```
Frame 304887: 146 bytes on wire (1168 bits), 146 bytes captured (1168 bits) on
Ethernet II, Src: Raspberr_ca:1f:1e (b8:27:eb:ca:1f:1e), Dst: AsrockIn_29:d3:3e
    Destination: AsrockIn_29:d3:3e (d0:50:99:29:d3:3e)
    Source: Raspberr_ca:1f:1e (b8:27:eb:ca:1f:1e)
    Type: IPv4 (0x0800)
Internet Protocol Version 4, Src: 10.1.1.31, Dst: 10.1.1.2
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes
    Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 132
    Identification: 0x19d0 (6608)
    Flags: 0x00
    Fragment offset: 0
    Time to live: 64
    Protocol: UDP (17)
    Header checksum: 0x4a77 [validation disabled]
    Source: 10.1.1.31
    Destination: 10.1.1.2
    [Source GeoIP: Unknown]
    [Destination GeoIP: Unknown]
User Datagram Protocol, Src Port: 53 (53), Dst Port: 63934 (63934)
    Source Port: 53
    Destination Port: 63934
    Length: 112
    Checksum: 0xceb2 [validation disabled]
    [Stream index: 340]
Domain Name System (response)
    [Request In: 300358]
    [Time: 1.948991000 seconds]
    Transaction ID: 0x0013
    Flags: 0x8180 Standard query response, No error
    Questions: 1
    Answer RRs: 1
    Authority RRs: 1
    Additional RRs: 0
    Queries
        mail.cs.up.ac.za: type AAAA, class IN
            Name: mail.cs.up.ac.za
            [Name Length: 16]
            [Label Count: 5]
            Type: AAAA (IPv6 Address) (28)
            Class: IN (0x0001)
    Answers
        mail.cs.up.ac.za: type CNAME, class IN, cname hermes.cs.up.ac.za
    Authoritative nameservers
        cs.up.ac.za: type SOA, class IN, mname ns2.up.ac.za
```

x

Packets for Question 5

Packet A

```
Frame 549655: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on inte
Ethernet II, Src: Giga-Byt_e9:68:40 (74:d4:35:e9:68:40), Dst: Tp-LinkT_e7:76:4
Internet Protocol Version 4, Src: 10.1.1.15, Dst: 54.164.213.10
Transmission Control Protocol, Src Port: 52400 (52400), Dst Port: 80 (80), Seq:
    Source Port: 52400
    Destination Port: 80
    [Stream index: 252]
    [TCP Segment Len: 0]
    Sequence number: 0    (relative sequence number)
    Acknowledgment number: 0
    Header Length: 40 bytes
    Flags: 0x002 (SYN)
        000. .... .... = Reserved: Not set
        ...0 .... .... = Nonce: Not set
        .... 0... .... = Congestion Window Reduced (CWR): Not set
        .... .0.. .... = ECN-Echo: Not set
        .... ..0. .... = Urgent: Not set
        .... ...0 .... = Acknowledgment: Not set
        .... .... 0... = Push: Not set
        .... .... .0.. = Reset: Not set
        .... .... ..1. = Syn: Set
        .... .... ...0 = Fin: Not set
        [TCP Flags: **********S*]
    Window size value: 29200
    [Calculated window size: 29200]
    Checksum: 0x827b [validation disabled]
    Urgent pointer: 0
    Options: (20 bytes), Maximum segment size, SACK permitted, Timestamps, No-
```

Y

Packet B

```
Frame 550473: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on inte
Ethernet II, Src: Tp-LinkT_e7:76:46 (30:b5:c2:e7:76:46), Dst: Giga-Byt_e9:68:4(
Internet Protocol Version 4, Src: 54.164.213.10, Dst: 10.1.1.15
Transmission Control Protocol, Src Port: 80 (80), Dst Port: 52400 (52400), Seq:
    Source Port: 80
    Destination Port: 52400
    [Stream index: 252]
    [TCP Segment Len: 0]
    Sequence number: 0    (relative sequence number)
    Acknowledgment number: 1    (relative ack number)
    Header Length: 40 bytes
    Flags: 0x012 (SYN, ACK)
        000. .... .... = Reserved: Not set
        ...0 .... .... = Nonce: Not set
        .... 0... .... = Congestion Window Reduced (CWR): Not set
        .... .0.. .... = ECN-Echo: Not set
        .... ..0. .... = Urgent: Not set
        .... ...1 .... = Acknowledgment: Set
        .... .... 0... = Push: Not set
        .... .... .0.. = Reset: Not set
        .... .... ..1. = Syn: Set
        .... .... ...0 = Fin: Not set
        [TCP Flags: *******A**S*]
    Window size value: 17898
    [Calculated window size: 17898]
    Checksum: 0x01c2 [validation disabled]
    Urgent pointer: 0
    Options: (20 bytes), Maximum segment size, SACK permitted, Timestamps, No-(
```

Packet C

```
Frame 550475: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on inte
Ethernet II, Src: Giga-Byt_e9:68:40 (74:d4:35:e9:68:40), Dst: Tp-LinkT_e7:76:4(
Internet Protocol Version 4, Src: 10.1.1.15, Dst: 54.164.213.10
Transmission Control Protocol, Src Port: 52400 (52400), Dst Port: 80 (80), Seq:
    Source Port: 52400
    Destination Port: 80
    [Stream index: 252]
    [TCP Segment Len: 0]
    Sequence number: 1    (relative sequence number)
    Acknowledgment number: 1    (relative ack number)
    Header Length: 32 bytes
    Flags: 0x010 (ACK)
        000. .... .... = Reserved: Not set
        ...0 .... .... = Nonce: Not set
        .... 0... .... = Congestion Window Reduced (CWR): Not set
        .... .0.. .... = ECN-Echo: Not set
        .... ..0. .... = Urgent: Not set
        .... ...1 .... = Acknowledgment: Set
        .... .... 0... = Push: Not set
        .... .... .0.. = Reset: Not set
        .... .... ..0. = Syn: Not set
        .... .... ...0 = Fin: Not set
        [TCP Flags: *******A****]
    Window size value: 229
    [Calculated window size: 29312]
    [Window size scaling factor: 128]
    Checksum: 0x7542 [validation disabled]
    Urgent pointer: 0
    Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
```

Packet D

```
Frame 550478: 683 bytes on wire (5464 bits), 683 bytes captured (5464 bits) on
Ethernet II, Src: Giga-Byt_e9:68:40 (74:d4:35:e9:68:40), Dst: Tp-LinkT_e7:76:4(
Internet Protocol Version 4, Src: 10.1.1.15, Dst: 54.164.213.10
Transmission Control Protocol, Src Port: 52400 (52400), Dst Port: 80 (80), Seq:
    Source Port: 52400
    Destination Port: 80
    [Stream index: 252]
    [TCP Segment Len: 617]
    Sequence number: 1    (relative sequence number)
    [Next sequence number: 618    (relative sequence number)]
    Acknowledgment number: 1    (relative ack number)
    Header Length: 32 bytes
    Flags: 0x018 (PSH, ACK)
        000. .... .... = Reserved: Not set
        ...0 .... .... = Nonce: Not set
        .... 0... .... = Congestion Window Reduced (CWR): Not set
        .... .0.. .... = ECN-Echo: Not set
        .... ..0. .... = Urgent: Not set
        .... ...1 .... = Acknowledgment: Set
        .... .... 1... = Push: Set
        .... .... .0.. = Reset: Not set
        .... .... ..0. = Syn: Not set
        .... .... ...0 = Fin: Not set
        [TCP Flags: *******AP***]
    Window size value: 229
    [Calculated window size: 29312]
    [Window size scaling factor: 128]
    Checksum: 0x5c86 [validation disabled]
    Urgent pointer: 0
    Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
Hypertext Transfer Protocol
    GET /livecountping/142650410/630826641504?__=ghidbpv89h2c&routed=1&jid&site
    Host: 1c48.dsr.livefyre.com\r\n
    User-Agent: Mozilla/5.0 (Windows NT 10.0) AppleWebKit/537.36 (KHTML, like (
    Origin: http://www.thelancet.com\r\n
    Accept: */*\r\n
    Referer: http://www.thelancet.com/journals/laninf/article/PIIS1473-3099(14)
    Accept-Encoding: gzip, deflate, sdch\r\n
    Accept-Language: en-GB,en;q=0.8,en-US;q=0.6,af;q=0.4,en-ZA;q=0.2\r\n
    Via: 1.1 styx (squid/3.3.8)\r\n
    X-Forwarded-For: 10.1.1.23\r\n
    Cache-Control: max-age=0\r\n
    Connection: keep-alive\r\n
    \r\n
```

Packet E

```
Frame 551367: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on int
Ethernet II, Src: Tp-LinkT_e7:76:46 (30:b5:c2:e7:76:46), Dst: Giga-Byt_e9:68:4
Internet Protocol Version 4, Src: 54.164.213.10, Dst: 10.1.1.15
Transmission Control Protocol, Src Port: 80 (80), Dst Port: 52400 (52400), Seq:
    Source Port: 80
    Destination Port: 52400
    [Stream index: 252]
    [TCP Segment Len: 0]
    Sequence number: 1    (relative sequence number)
    Acknowledgment number: 618    (relative ack number)
    Header Length: 32 bytes
    Flags: 0x010 (ACK)
        000. .... .... = Reserved: Not set
        ...0 .... .... = Nonce: Not set
        .... 0... .... = Congestion Window Reduced (CWR): Not set
        .... .0.. .... = ECN-Echo: Not set
        .... ..0. .... = Urgent: Not set
        .... ...1 .... = Acknowledgment: Set
        .... .... 0... = Push: Not set
        .... .... .0.. = Reset: Not set
        .... .... ..0. = Syn: Not set
        .... .... ...0 = Fin: Not set
        [TCP Flags: *******A****]
    Window size value: 31
    [Calculated window size: 15872]
    [Window size scaling factor: 512]
    Checksum: 0x7327 [validation disabled]
    Urgent pointer: 0
    Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
```

Packet F

```
1 Frame 551378: 657 bytes on wire (5256 bits), 657 bytes captured (5256 bits) on
2 Ethernet II, Src: Tp-LinkT_e7:76:46 (30:b5:c2:e7:76:46), Dst: Giga-Byt_e9:68:4(
3 Internet Protocol Version 4, Src: 54.164.213.10, Dst: 10.1.1.15
4 Transmission Control Protocol, Src Port: 80 (80), Dst Port: 52400 (52400), Seq:
5       Source Port: 80
6       Destination Port: 52400
7       [Stream index: 252]
8       [TCP Segment Len: 591]
9       Sequence number: 1    (relative sequence number)
10      [Next sequence number: 592    (relative sequence number)]
11      Acknowledgment number: 618    (relative ack number)
12      Header Length: 32 bytes
13      Flags: 0x018 (PSH, ACK)
14           000. .... .... = Reserved: Not set
15           ...0 .... .... = Nonce: Not set
16           .... 0... .... = Congestion Window Reduced (CWR): Not set
17           .... .0.. .... = ECN-Echo: Not set
18           .... ..0. .... = Urgent: Not set
19           .... ...1 .... = Acknowledgment: Set
20           .... .... 1... = Push: Set
21           .... .... .0.. = Reset: Not set
22           .... .... ..0. = Syn: Not set
23           .... .... ...0 = Fin: Not set
24           [TCP Flags: *******AP***]
25      Window size value: 31
26      [Calculated window size: 15872]
27      [Window size scaling factor: 512]
28      Checksum: 0x8e38 [validation disabled]
29      Urgent pointer: 0
30      Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
32 Hypertext Transfer Protocol
33      HTTP/1.1 200 OK\r\n
34      Server: Livecount/1.1 (Simple 4.0) lc48.dsr.livefyre.com\r\n
37      Access-Control-Allow-Origin: http://www.thelancet.com\r\n
38      Access-Control-Max-Age: 1728000\r\n
39      Access-Control-Allow-Methods: GET, OPTIONS\r\n
40      Access-Control-Allow-Headers: Origin, Accept-Language, Accept-Encoding, X-F
41      Access-Control-Allow-Credentials: true\r\n
42      Connection: close\r\n
43      Content-Type: application/json\r\n
44      \r\n
```

Packet G

```
Frame 551380: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on int
Ethernet II, Src: Giga-Byt_e9:68:40 (74:d4:35:e9:68:40), Dst: Tp-LinkT_e7:76:4
Internet Protocol Version 4, Src: 10.1.1.15, Dst: 54.164.213.10
Transmission Control Protocol, Src Port: 52400 (52400), Dst Port: 80 (80), Seq:
    Source Port: 52400
    Destination Port: 80
    [Stream index: 252]
    [TCP Segment Len: 0]
    Sequence number: 618    (relative sequence number)
    Acknowledgment number: 592    (relative ack number)
    Header Length: 32 bytes
    Flags: 0x010 (ACK)
        000. .... .... = Reserved: Not set
        ...0 .... .... = Nonce: Not set
        .... 0... .... = Congestion Window Reduced (CWR): Not set
        .... .0.. .... = ECN-Echo: Not set
        .... ..0. .... = Urgent: Not set
        .... ...1 .... = Acknowledgment: Set
        .... .... 0... = Push: Not set
        .... .... .0.. = Reset: Not set
        .... .... ..0. = Syn: Not set
        .... .... ...0 = Fin: Not set
        [TCP Flags: *******A****]
    Window size value: 238
    [Calculated window size: 30464]
    [Window size scaling factor: 128]
    Checksum: 0x6fa8 [validation disabled]
    Urgent pointer: 0
    Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
```

Packet H

```
Frame 552818: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on inte
Ethernet II, Src: Tp-LinkT_e7:76:46 (30:b5:c2:e7:76:46), Dst: Giga-Byt_e9:68:4(
Internet Protocol Version 4, Src: 54.164.213.10, Dst: 10.1.1.15
Transmission Control Protocol, Src Port: 80 (80), Dst Port: 52400 (52400), Seq:
    Source Port: 80
    Destination Port: 52400
    [Stream index: 252]
    [TCP Segment Len: 0]
    Sequence number: 592    (relative sequence number)
    Acknowledgment number: 618    (relative ack number)
    Header Length: 32 bytes
    Flags: 0x011 (FIN, ACK)
        000. .... .... = Reserved: Not set
        ...0 .... .... = Nonce: Not set
        .... 0... .... = Congestion Window Reduced (CWR): Not set
        .... .0.. .... = ECN-Echo: Not set
        .... ..0. .... = Urgent: Not set
        .... ...1 .... = Acknowledgment: Set
        .... .... 0... = Push: Not set
        .... .... .0.. = Reset: Not set
        .... .... ..0. = Syn: Not set
        .... .... ...1 = Fin: Set
        [TCP Flags: *******A***F]
    Window size value: 31
    [Calculated window size: 15872]
    [Window size scaling factor: 512]
    Checksum: 0x6eae [validation disabled]
    Urgent pointer: 0
    Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
```

Packet I

```
Frame 552820: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on int
Ethernet II, Src: Giga-Byt_e9:68:40 (74:d4:35:e9:68:40), Dst: Tp-LinkT_e7:76:4
Internet Protocol Version 4, Src: 10.1.1.15, Dst: 54.164.213.10
Transmission Control Protocol, Src Port: 52400 (52400), Dst Port: 80 (80), Seq:
    Source Port: 52400
    Destination Port: 80
    [Stream index: 252]
    [TCP Segment Len: 0]
    Sequence number: 618    (relative sequence number)
    Acknowledgment number: 593    (relative ack number)
    Header Length: 32 bytes
    Flags: 0x011 (FIN, ACK)
        000. .... .... = Reserved: Not set
        ...0 .... .... = Nonce: Not set
        .... 0... .... = Congestion Window Reduced (CWR): Not set
        .... .0.. .... = ECN-Echo: Not set
        .... ..0. .... = Urgent: Not set
        .... ...1 .... = Acknowledgment: Set
        .... .... 0... = Push: Not set
        .... .... .0.. = Reset: Not set
        .... .... ..0. = Syn: Not set
        .... .... ...1 = Fin: Set
        [TCP Flags: *******A***F]
    Window size value: 238
    [Calculated window size: 30464]
    [Window size scaling factor: 128]
    Checksum: 0x6c2f [validation disabled]
    Urgent pointer: 0
    Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
```

Packet J

```
Frame 552843: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on int
Ethernet II, Src: Tp-LinkT_e7:76:46 (30:b5:c2:e7:76:46), Dst: Giga-Byt_e9:68:4
Internet Protocol Version 4, Src: 54.164.213.10, Dst: 10.1.1.15
Transmission Control Protocol, Src Port: 80 (80), Dst Port: 52400 (52400), Seq:
    Source Port: 80
    Destination Port: 52400
    [Stream index: 252]
    [TCP Segment Len: 0]
    Sequence number: 593    (relative sequence number)
    Acknowledgment number: 619    (relative ack number)
    Header Length: 32 bytes
    Flags: 0x010 (ACK)
        000. .... .... = Reserved: Not set
        ...0 .... .... = Nonce: Not set
        .... 0... .... = Congestion Window Reduced (CWR): Not set
        .... .0.. .... = ECN-Echo: Not set
        .... ..0. .... = Urgent: Not set
        .... ...1 .... = Acknowledgment: Set
        .... .... 0... = Push: Not set
        .... .... .0.. = Reset: Not set
        .... .... ..0. = Syn: Not set
        .... .... ...0 = Fin: Not set
        [TCP Flags: *******A****]
    Window size value: 31
    [Calculated window size: 15872]
    [Window size scaling factor: 512]
    Checksum: 0x6cbc [validation disabled]
    Urgent pointer: 0
```

Packets for Question 4

Packet 1

```
Frame 297947: 76 bytes on wire (608 bits), 76 bytes captured (608 bits) on inte
Ethernet II, Src: AsrockIn_29:d3:3e (d0:50:99:29:d3:3e), Dst: Raspberr_ca:1f:1e
    Destination: Raspberr_ca:1f:1e (b8:27:eb:ca:1f:1e)
    Source: AsrockIn_29:d3:3e (d0:50:99:29:d3:3e)
    Type: IPv4 (0x0800)
Internet Protocol Version 4, Src: 10.1.1.2, Dst: 10.1.1.31
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes
    Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 62
    Identification: 0x33e8 (13288)
    Flags: 0x00
    Fragment offset: 0
    Time to live: 128
    Protocol: UDP (17)
    Header checksum: 0xf0a4 [validation disabled]
    Source: 10.1.1.2
    Destination: 10.1.1.31
    [Source GeoIP: Unknown]
    [Destination GeoIP: Unknown]
User Datagram Protocol, Src Port: 63933 (63933), Dst Port: 53 (53)
    Source Port: 63933
    Destination Port: 53
    Length: 42
    Checksum: 0x81ce [validation disabled]
    [Stream index: 337]
Domain Name System (query)
    Transaction ID: 0x0012
    Flags: 0x0100 Standard query
    Questions: 1
    Answer RRs: 0
    Authority RRs: 0
    Additional RRs: 0
    Queries
        mail.cs.up.ac.za: type A, class IN
            Name: mail.cs.up.ac.za
            [Name Length: 16]
            [Label Count: 5]
            Type: A (Host Address) (1)
            Class: IN (0x0001)
```

Packet 2

```
Frame 297957: 87 bytes on wire (696 bits), 87 bytes captured (696 bits) on int
Ethernet II, Src: Raspberr_ca:1f:1e (b8:27:eb:ca:1f:1e), Dst: Tp-LinkT_e7:76:4
    Destination: Tp-LinkT_e7:76:46 (30:b5:c2:e7:76:46)
    Source: Raspberr_ca:1f:1e (b8:27:eb:ca:1f:1e)
    Type: IPv4 (0x0800)
Internet Protocol Version 4, Src: 10.1.1.31, Dst: 8.8.8.8
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes
    Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 73
    Identification: 0xbee2 (48866)
    Flags: 0x00
    Fragment offset: 0
    Time to live: 64
    Protocol: UDP (17)
    Header checksum: 0xa092 [validation disabled]
    Source: 10.1.1.31
    Destination: 8.8.8.8
    [Source GeoIP: Unknown]
    [Destination GeoIP: Unknown]
User Datagram Protocol, Src Port: 8312 (8312), Dst Port: 53 (53)
    Source Port: 8312
    Destination Port: 53
    Length: 53
    Checksum: 0xabb1 [validation disabled]
    [Stream index: 338]
Domain Name System (query)
    Transaction ID: 0x80b0
    Flags: 0x0110 Standard query
    Questions: 1
    Answer RRs: 0
    Authority RRs: 0
    Additional RRs: 1
    Queries
        mail.cs.up.ac.za: type A, class IN
            Name: mail.cs.up.ac.za
            [Name Length: 16]
            [Label Count: 5]
            Type: A (Host Address) (1)
            Class: IN (0x0001)
```

COURSE: Computer Science
PAPER: COS332

Question 1

In each case select the alternative that fits the question best and write only the corresponding letter on your answer sheet.

1) Electromagnetic interference lasting 0,02 seconds affects data being transmitted via a 33600 bps line. How many bits may have changed?

   A: ±1 680 000

   B: ±670

   C: ±33 600

   D: 0

   E: ±84

0,02

1/50 × 33 600

50 ×
670
33500

2) A switch (used in the place of a hub)

   A: Connects different networks and can convert protocols.

   B: Makes a connection only between computers that are busy communicating.

   C: Routes packets between nodes.

   D: Operates on ISO OSI layer 3.

   E: More than one of the above

3) Which of the following is (are) *not* a function of ISO/OSI's layer 2?

   A: Media access control

   B: Data delineation

   C: Dialogue control

   D: Error control

   E: More than one of the above

4) Which of the following do/does *not* include contention control?

A: UTP

B: Token ring

C: CSMA/CD

D: Ethernet

E: More than one of the above

5) A header added to a message at ISO OSI layer n will be removed at the destination at layer ...

A: $n - 1$

B: n

C: $n + 1$

D: 1

E: 7

6) Layers 1 to 4 of the ISO OSI model are known as the

A: Network-oriented layers

B: Application-oriented layers

C: TCP/IP-oriented layers

D: More than one of the above

E: None of the above

7) By whom are technical aspects of the Internet standardised?

A: IETF

B: ISO

C: EIA

D: ITU

E: CCITT

4/...

8) What is specified by X.500?

A: E-mail

B: Directory services

C: Packet networks

D: Telephone services

E: None of the above

9) Which of the following is *not* a valid DNS record type?

A: PTR

B: MX

C: AAAA

D: AA

E: A

10) Which of the following is *not* a valid `nslookup` command?

A: `ls example.com`

B: `set type=any`

C: `server 8.8.8.8`

D: `exit`

E: `resolve up.ac.za`

11) The structure used to store data used by SNMP is known as ...

A: DSNMP

B: X.509

C: MIB

D: NMP

E: NIC

12) Assume you are writing a server that is intended to be accessed by a Telnet client.

A: Normally the server should echo every character received, because the client may not be able to echo typed characters locally.

B: The server should not echo received input because echoing will cause characters to be displayed twice on modern Telnet clients.

C: The server should query the Telnet client to determine whether it echoes locally; the server should then only echo if the client will not echo.

D: The server should issue an ANSI escape sequence to disable local echoing at the client; the server should then echo everything received.

E: A good compromise is to echo every second character received; the user should be still be able to determine the intended output even if some characters are missing and others are repeated.

13) Which of the following is *not* an SNMP request?

A: `Get`

B: `GetNext`

C: `GetBulk`

D: `Reboot`

E: `Set`

14) When one node polls another, it wants to know whether

A: The other node has something to send.

B: The other node is communicating.

C: It may send something to the other node.

D: None of the above

15) A *zone transfer* refers to the process of

A: Moving hosting from one service provider to another.

B: Selling a domain name to someone else.

C: Copying DNS entries from a primary server to a secondary server.

D: Accessing data outside the corporate network, or data in the demilitarised zone of the corporate network.

E: Obtaining new name servers.

16) When no monitor's presence is detected on an 802.5 network

A: The hub appoints a new monitor.

B: The previous monitor resumes its duties.

C: The station with the highest priority starts acting as monitor.

D: A new monitor is elected by the remaining nodes.

E: The network simply continues operating because it does not really need a monitor.

17) Which layer operates in terms of *frames*?

A: Physical

B: Data link

C: Transport

D: Network

E: Application

18) IEEE 802.11 standardises

A: Ethernet

B: Token bus

C: Token ring

D: Security

E: Wireless LANs

19) When a token is lost in an IEEE 802.5 network, the network uses the following algorithm to regenerate the token:

A: Bully

B: Largest

C: Friend

D: Sweet

E: None of the above

20) *Hop count* refers to

A: The number of times a packet has been bounced between a client and a server.

B: A routing metric.

C: The number of frames that have been dropped.

D: The number of times a packet has been bounced between two routers.

E: The number of times an RFC has been revised.

21) An IPv4 multicast packet

A: Should be delivered to all subscribed parties.

B: In TCP/IP has an IP address with a first byte between 224 and 239, both included.

C: Is routed like a class C packet.

D: More than one of the above

E: All of the above

22) ARP translates a(n) ... address to a(n) ... address.

A: IP, MAC
B: MAC, IP
C: TCP, IP
D: IPv4, IPv6
E: MAC, CAM

23) Assume a HTTP request includes the following:
`Accept-Charset: iso-8859-1, utf-8;q=0.5, *;q=0.7`
Suppose further that the server supports (only) UTF-8 and UTF-16. Which code will be used?

A: ISO-8559-1
B: UTF-8
C: UTF-16
D: ASCII
E: None of the above

24) On which layer is SSL/TLS usually positioned?

A: 6
B: 5
C: 'Just above 5'
D: 'Just above 4'
E: 4

25) EBNF is

A: An abbreviation for Extended Backus-Naur Form
B: Often used by CFPs to specify the content of data messages
C: Standardised by a CFP
D: More than one of the above
✓ **E:** All of the above

26) An email message that contains a PDF document as an attachment may use the following MIME subtype to encapsulate the mail message and its attachment.

A: `multipart/mixed`

B: `multipart/digest`

C: `multipart/alternative`

D: `multipart/plain`

E: `text/plain`

27) To use POP3 anonymously, one uses the command

A: USER ANON

B: USER ANONYMOUS

C: NOUSER

D: USER *

E: None of the above

28) The TTL field in an IP datagram

A: Is used to encode a PING message.

B: Determines the destination port of the datagram.

C: Indicates whether the datagram has been fragmented.

D: Determines the maximum number of hops over which the packet can be forwarded.

E: None of the above

29) IP datagrams support source routing. Source routing is

A: Where the packet is routed back to the source.

B: Where all the packets from the same source are all sent together.

C: Where the source determines the route of the packet.

D: Frequently used on the Internet.

E: More than one of the above

16 ×
8

128

30) An IPv6 address consists of

A: 128 bits.

B: 16 octets.

C: 8 hexadecimal numbers, separated by colons.

✓ D: More than one of the above

E: All of the above

31) ICMP is used to

A: Report errors that occur on the IP layer.

B: Test network functions.

C: Modify routes.

D: More than one of the above

E: All of the above

32) Where are fragmented IP datagrams reassembled?

A: At the first router that notices that it is fragmented.

B: At the first router that can handle the total IP datagram.

C: At the final destination.

D: Whenever the TTL becomes 0.

E: Fragmented IP datagrams are discarded, rather than reassembled.

33) An ARP cache is maintained on

A: Each host connected to an Ethernet bus.

B: A special ARP cache server.

C: Layer 4 of the protocol stack.

D: The RARP server.

E: The hub.

34) A typical ARP message looks as follows:

A: Who has 10.1.1.1? Tell 10.1.1.2.

B: Who has 00-04-38-68-da-00? Tell 00-11-43-e1-ba-4d.

C: `GET / HTTP/1.1`

D: What is the IP of xx.co.za?

E: What is the name of 196.7.147.37?

35) An AS (Autonomous System) is usually a

A: Personal computer.

B: Network controlled by a single authority.

C: Mainframe computer.

D: ccTLD.

E: DNS registrar.

36) Suppose an entry in node A's routing table indicates that node B is the next node on the path to Z. Suppose an entry in node B's routing table indicates that the next hop on the path to Z is A. The resulting problem is addressed by using the following IP header field:

A: TTL

B: Source address

C: Destination address

D: Network unreachable

E: Been-there flag

37) IP packets may be fragmented because of

A: Limited size of TCP packets

B: Maximum PDU size

C: Congestion

D: Number of hops between source and destination

E: Limited bandwidth

38) If a TCP segment is lost, TCP reacts by

A: Retransmitting the segment.

B: Decreasing the rate at which segments are sent.

C: Performing another three-way handshake.

D: More than one of the above

E: All of the above

39) A half-open attack occurs when

A: Only the first step of a three-way handshake is performed.

B: Only the first two steps of a three-way handshake are performed.

C: All three steps of a three-way handshake are performed.

D: A four-way handshake is performed.

E: No handshake is performed.

40) Which protocol has reliability as one of its main goals?

A: IP

B: TCP

C: UDP

D: More than one of the above

E: All of the above

41) Which field does *not* occur in the UDP header?

A: Source port

B: Destination port

C: Sequence number

D: Length

E: Checksum

42) TFTP

A: Has strong access control.
B: Uses UDP.
C: Cannot be used with DNS.
D: More than one of the above
E: All of the above

43) The following network protocol(s) is (are) used for network management:

A: SNMP
B: CMIP
C: CMOT
D: More than one of the above
E: All of the above

44) Which of the following messages can be used in SMTP?

A: HELO
B: SENDER
C: MESS
D: EHLO
E: More than one of the above

45) Which of the following statements regarding DHCP is false?

A: DHCP was designed as an extension of BOOTP.
B: DHCP can supply the address of the default router.
C: DHCP uses IP to carry data.
D: DHCP will not assign an address assigned to computer A to a computer B before A has been been switched of.
E: DHCP uses UDP to carry data.

46) If SMTP cannot deliver a message, it may

A: Wait and try again later.

B: Send the message to a secondary server.

C: Return an error message to the sender.

D: More than one of the above

E: All of the above

47) The software providing services on the internet is often known as a

A: Dæmon

B: Spirit

C: Ghost

D: Waiter

E: Client

48) To express the TLD `com` as an FQDN one would write

A: `.com`

B: `com.`

C: `com.arpa`

D: `[com]`

E: `*.com`

49) In the DNS, SOA refers to

A: Service-oriented Architecture

B: IPv6

C: Start of Authority

D: Start of Address

E: Start of Aliases

50) The primary reason for including the `host:` header in the HTTP protocol is to

A: Ensure that the `hosts` file cannot be misused.

B: Make a clear distinction between client and host requests.

C: Enable virtual hosting.

D: Enhance security by ensuring that the destination server is indeed the one the user wanted to visit because the name of the server can now be compared with the name provided in the `host:` header.

E: Specify alternative hosts that may contain the content the requester wants.

[50]

Question 2

With which well-known ports are the following services usually associated?

a) *HTTP*

b) *DNS*

c) *SMTP*

d) *Telnet*

e) *SSH*

[5]

16./...

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0 10 110 1110

Question 3

The various UTF encoding schemes work by arranging the Unicode character set into a sequence where frequently used characters are associated with small numbers and less frequently used characters with larger numbers. These 'sequence' numbers are mapped to the 'real' Unicode representation.

Consider the following (hexadecimal) sequence of characters represented in UTF-8:

```
C3 8B E6 81 99 CC A5 77
```

a) How many Unicode characters occur in this sequence? (1)

b) What are the UTF-8 'sequence' numbers of each of these characters? Write each sequence number on a new line. Write each of the numbers in hexadecimal. (8)

c) Assume you encounter the following sequences of bytes in what should be a string in UTF-8:

```
46 A7 20 17 66
```

0 10 110 1110

This sequence cannot be correct. At which byte will the decoder first notice a problem? Provide the value of this byte, rather than its position in the sequence. (1)

ABCDEF
10 11 12 13 14 15

[10]

17./...

3 8 B E 6 8 1

0 0011 1000 1011 11100110 1000 0001

4 2

9 C C A 5 7 7

001 1100 1110 1010 0101 0111 0111

3 1

6 A 7

0 0110 1010 0111

Question 4

Assume that computers on an internet are addressed using single byte addresses expressed in decimal. An address will therefore simply be a number, such as 1, 10 or 254. In all other respects the protocol behaves identical to IPv4. Assume node 1 is a root name server and you use node 100 as your default name server to perform *recursive* name resolution. The following are (partial) zone files at some nodes running *non-recursive* DNS servers. The files are incomplete since some rows and columns have been omitted; these omissions do not affect the answers to the questions. No caching occurs.

Node 1		
com.	NS	10
org.	NS	11
mil.	NS	12
za.	NS	13

Node 13		
ac	NS	20
org	NS	21
co	NS	22
net	NS	23

Node 21		
xx	NS	47
yy	NS	48
zz	NS	49

Node 22		
xx	NS	33
yy	NS	55
zz	NS	34

Node 33		
@	NS	33
@	MX,1	98
www	A	40
ftp	A	42
mail	A	44
arthur	A	45
guinevere	NS	46

Node 46		
@	NS	46
@	A	60
www	A	61
ftp	A	62
mail	A	63
arthur	A	64

Node 47		
@	NS	47
@	MX,1	99
www	A	55
ftp	A	56
mail	A	57
arthur	A	58
guinevere	NS	59

The following series of `nslookup` queries is executed in sequence from top to bottom. Provide the address (that is, the number) returned by each of the queries. The queries are indicated by a Roman number in square brackets (such as "[iii]"). The commands that do not return responses are not numbered. Write `N/A` if the query cannot be resolved using the information provided above.

```
nslookup
> arthur.xx.org.za          [i]
> set type=NS
> com.                      [ii]
> co.za                     [iii]
> guinevere.xx.co.za        [iv]
> set type=A
> guinevere.xx.co.za        [v]
> set type=MX
> xx.co.za                  [vi]
> xx.org.za                 [vii]
> set type=A
> john.xx.org.za            [viii]
> server 1
> za.                       [ix]
> server 35
> www.xx.org.za             [x]
>
```

[10]

18./...

Question 5

Provide the (standard) IP routing algorithm using the scenario, functions and symbols as described below.

Assume an IP datagram I with destination address d arrives at a node N. Node N has IP address n. N has a routing table R. Assume R is function such that, for any network address x

$$R(x) = \begin{cases} h & \text{where } h \text{ is the designated next hop to } x \\ \lambda & \text{if no exlicit next hop is defined for } x \end{cases}$$

R contains no overlapping addresses, with one exception: Let ∞ be the network address `0.0.0.0/0`. Then $R(\infty)$ *may* return a host address. (If $R(\infty)$ is not defined to return a host address, it will return λ.) The netmask in use at N is m.

The bitwise AND operator is expressed as a single dot; the bitwise AND of two arbitrary binary strings p and q should therefore be written as $p.q$. The usual mathematical comparison tests ($=$, $\neq$, $<$, $\leq$, etc) may be used. Simple conditions may be combined using $\wedge$ (logical AND) and $\vee$ (logical OR). Parenthesis may be used if required.

The following actions are available for any datagram D and, where appropriate, any IP address y:

$D \rightarrow y$: Forward D to address y (via the data link layer)
$D\uparrow$: hand D to the higher layer of the protocol stack on the current machine
: (after removing the appropriate header information)
$D\downarrow$: hand D to the lower layer of the protocol stack on the current machine
: (after adding the appropriate header information)
$D\leftarrow$: return D to its sender
$D\dagger$: Drop D

Provide the five (5) steps of the routing algorithm that will be executed at N to route I. No ICMP transmissions should be included. Every step of your answer should start with one of the following words: `if`, `elseif` or `else`. Where appropriate (`if`, `elseif`) a logical test should be specified next, which should be followed by the word `then`. Finally an action should be specified for each step.

[5]

13
168

172.16.0.0 /21

511 255 −
19 120 7
248

1 0000.0000

31 218
. 0001,1111.11111,000 . 00000000
255 . 254

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Question 6

As you probably know, RFC1918 reserves the IPv4 address blocks 10.0.0.0/8, 172.16.0.0/12 and 192.168.0.0/16 (in the traditional A, B and C class address spaces, respectively) for private use. Assume ICANN decides that the private addresses in the class B network space are hardly ever used for their intended purpose, and decides to make these addresses available as public addresses. For technical reasons they decide to split the block into at least 500 blocks (or networks) such that each network can address at least 2000 hosts.

They therefore subnet the current private block into smaller public blocks (or networks) to achieve these goals. Let $\mathcal{A}$ be the new network with the smallest network address and $\mathcal{Z}$ be the new network with the largest network address. (Remember that they want *at least* 500 networks, but the scheme they use may result in more than 500 networks; similarly each network may support more than 2000 hosts.)

a) What is the network address of network $\mathcal{A}$? Use CIDR notation. (2)

b) What is the network address of network $\mathcal{Z}$? Use CIDR notation. (2)

c) What is the address range network $\mathcal{Z}$? In other words, what are the

(i) Smallest; and

(ii) Largest

IPv4 addresses that can be assigned to hosts in network $\mathcal{Z}$? (2)

d) What is the broadcast address for network $\mathcal{A}$? (1)

e) What is the broadcast address for network $\mathcal{Z}$? (1)

255 −
221
. 34

f) Consider a computer with the address 172.25.109.221.

(i) In which subnetwork will this computer be located? (Your answer will be a number from 0 to 499 — or more.)

(ii) What will host number will this computer have in this subnetwork? (Your answer will be a number from 1 to 2000 — or more.) (2)

128 + 192 + 64 +
64 32 32
192 224 96 +
8
104 +
4

[10]

0001.0000.00000,111. 255

20./...

25
221 + 256 + 0 + 1047

2, 0001,1001.01101,101.11011101

10010 1101

1 32 1 8 + 4 + 1
13
45

256 +
45
301

192 +
16
208 +
2

1048
256
221
1525

Scanned by CamScanner

Question 7

Client A is using TCP socket 44 to communicate TCP socket 55 on server B (where these ports are used for some non-standard purposes). After a while both A and B have thousands of bytes of data ready to send to the other computer. Both windows are currently full, so neither computer can transmit. Both computers will send as much data as possible during any given transmission, but neither computer will transmit more than a 1000 bytes in a single segment.

Normally A and B will communicate in full duplex. However, for the purposes of this question assume that messages are sent one at a time. Also assume that an action completes (for example, that a sent message arrives) before any subsequent action occurs.

The relevant events and counters are as follows:

I) The last byte A sent to B was byte 3344.
II) The last byte B sent to A was byte 4455.
III) B consumes 1500 bytes of data.
IV) B sends an acknowledgement to A.
V) A sends data to B.
VI) A sends data to B.
VII) A consumes 750 bytes of data.
VIII) B sends an acknowledgement to A.
IX) A sends an acknowledgement to B.
X) B sends data to A.

Your task is to provide the values of the following header fields of the transmitted segments listed above.

a) Source port of segment (IV)
b) Window advertisement of segment (IV)
c) Window advertisement of segment (V)
d) Length of segment (V)
e) Acknowledgement number of segment (VI)
f) Source port of segment (VIII)
g) Length of segment (VIII)
h) Window advertisement of segment (IX)
i) Destination port of segment (IX)
j) Acknowledgement number of segment (X)

[10]

TOTAL **[100]**

END OF PAPER